St. George's University
Grenada, West Indies

Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences | Lecture NB 15

Nasal Cavity

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Recommended reading

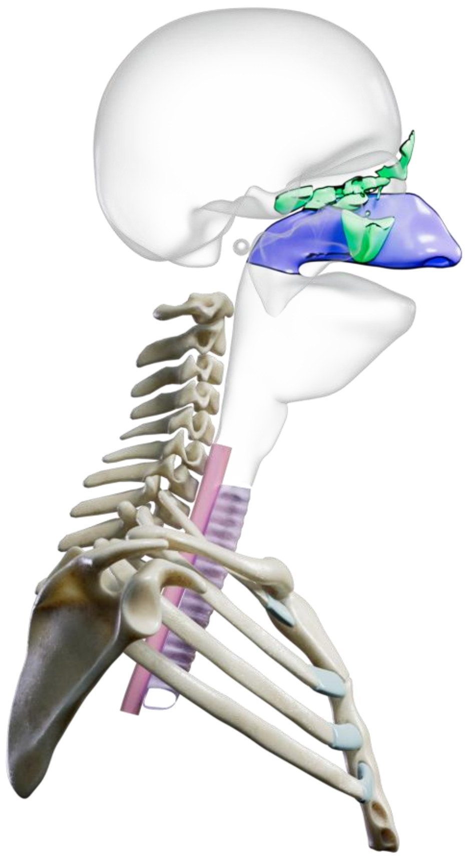
Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 5th ed. Elsevier. Chapter 8.

Pay special attention to the 'In the Clinic' boxes which contain the clinical correlates for this topic.

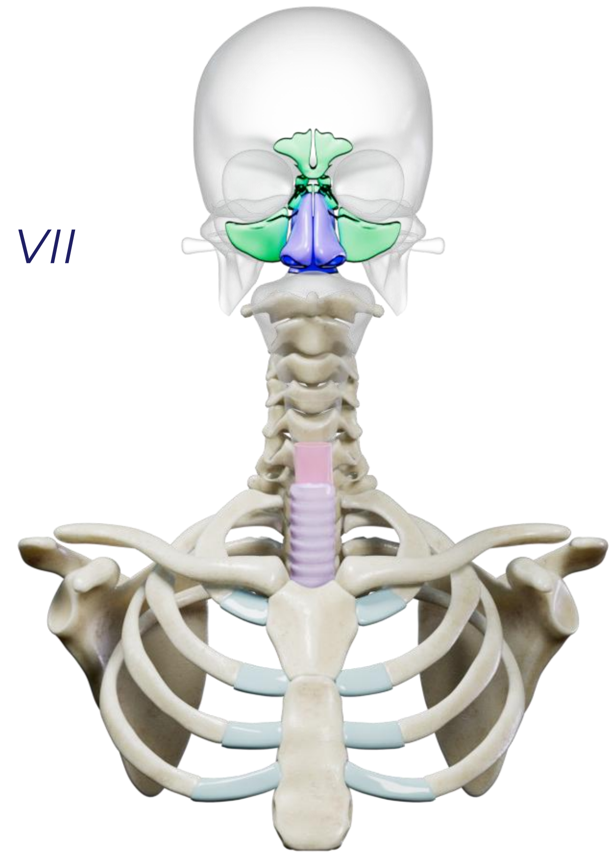
Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0096	List the bones that form the boundaries of the nasal cavity.
SOM.MKII.BPM2.1.NB.1.ANAT.0097	List the bones and cartilage that form the nasal septum.
SOM.MKII.BPM2.1.NB.1.ANAT.0098	Describe the sensory and parasympathetic innervation of the nasal cavity
SOM.MKII.BPM2.1.NB.1.ANAT.0099	Describe the structure and function of the turbinates
SOM.MKII.BPM2.1.NB.1.ANAT.0100	Describe the blood supply and lymphatic drainage of the nasal cavity
SOM.MKII.BPM2.1.NB.1.ANAT.0101	Describe the paranasal sinuses, the bones in which they are found and where each opens into the nasal cavity
SOM.MKII.BPM2.1.NB.1.ANAT.0102	Describe the sensory and parasympathetic innervations of the paranasal sinuses. (GAFS)
SOM.MKII.BPM2.1.NB.1.ANAT.0107	Explain the clinical significance of Kiesselbach’s area in relation to epistaxis
SOM.MKII.BPM2.1.NB.1.ANAT.0229	Identify major anatomical structures of the oral and nasal regions in MRI, CT and X-ray imaging

Overview



- External nose
- Nasal cavity
- Paranasal sinuses
- *Cranial Nerves: I, V, VII*

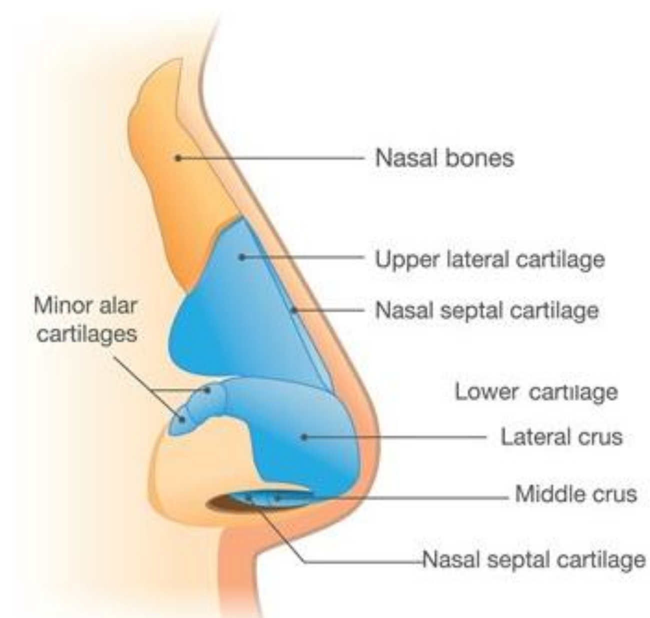


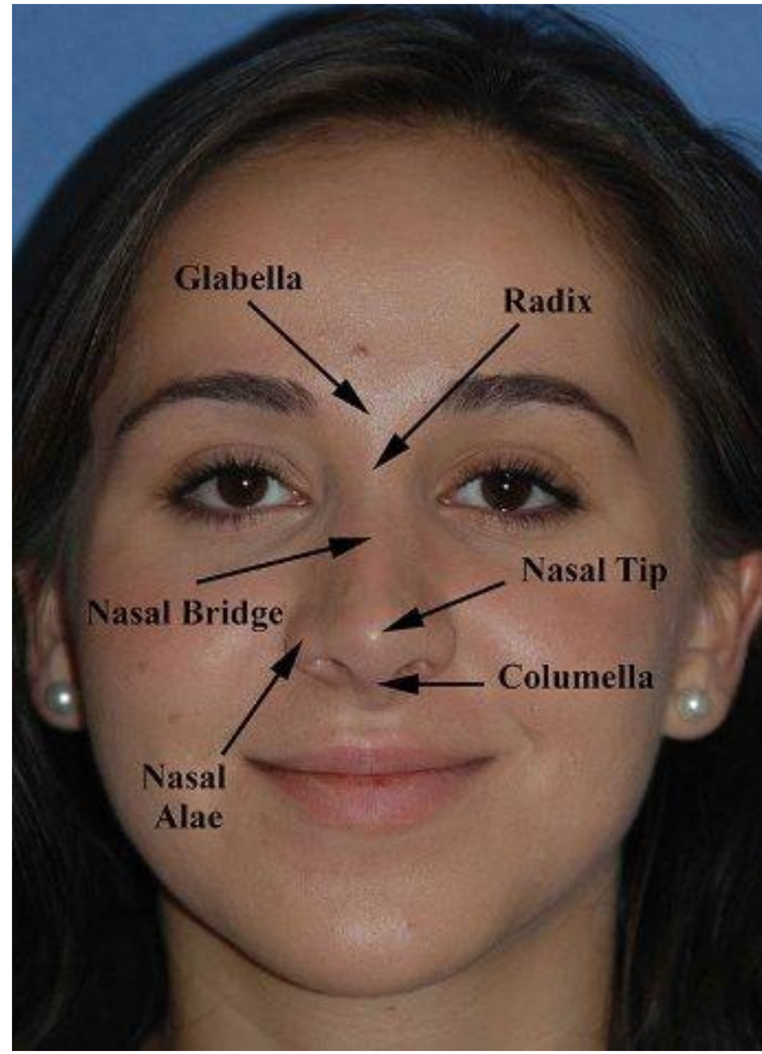
Nasal Cavity/ Nasopharynx

Sinuses

External Nose – Superficial Landmarks

- Visible projection from the face
- Made up of different **cartilages**
  - lateral
  - alar
  - septal

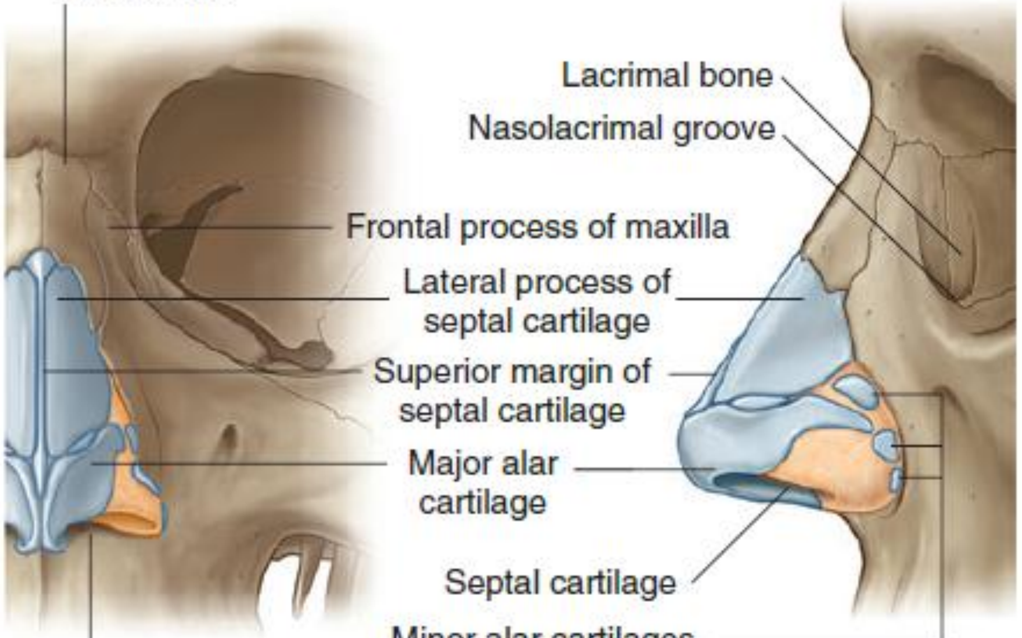




External Nose - Skeleton

Consists of

- Nasal bones
- Frontal processes of maxillae
- Nasal part of frontal bone and its nasal spine
- Bony parts of nasal septum

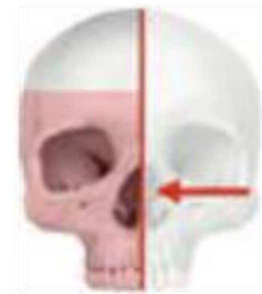


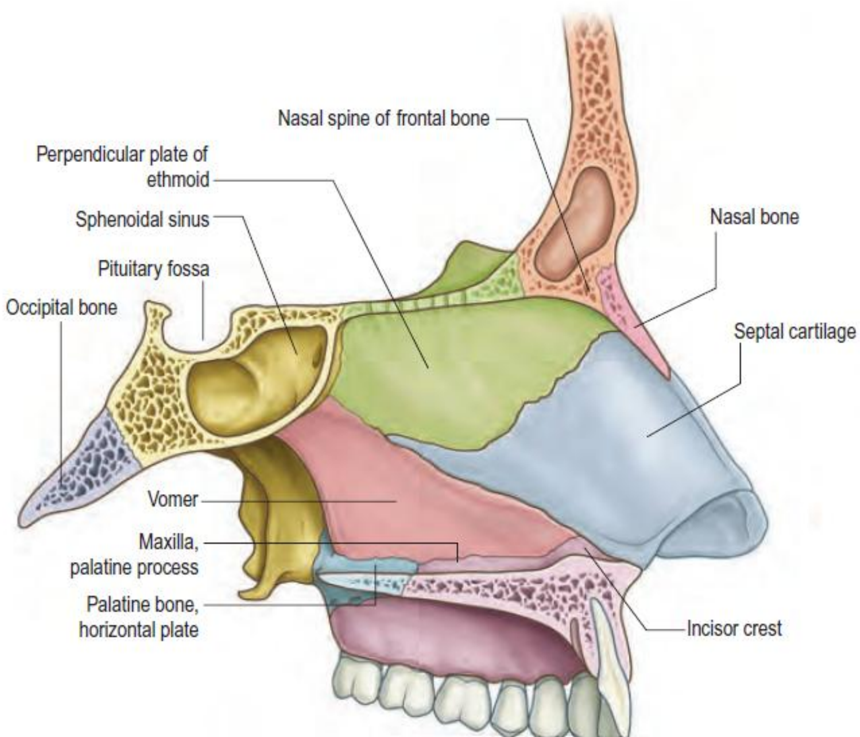
Nasal Septum

- Divides nasal chamber into two nasal cavities

Main components:

- *Cartilaginous part*
- *Bony part*
  - Vomer
  - Perpendicular plate of ethmoid





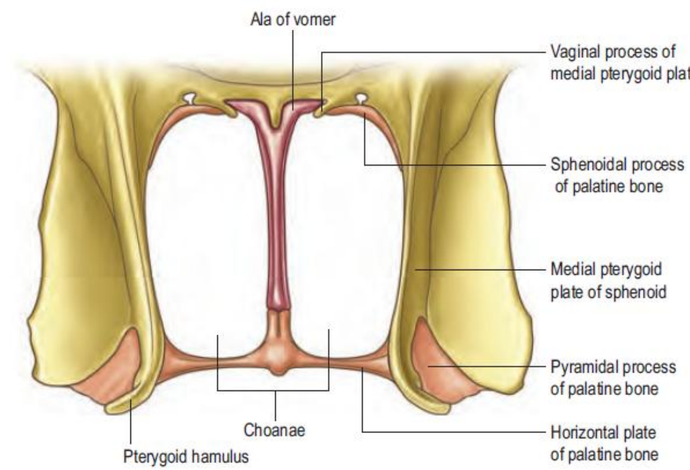
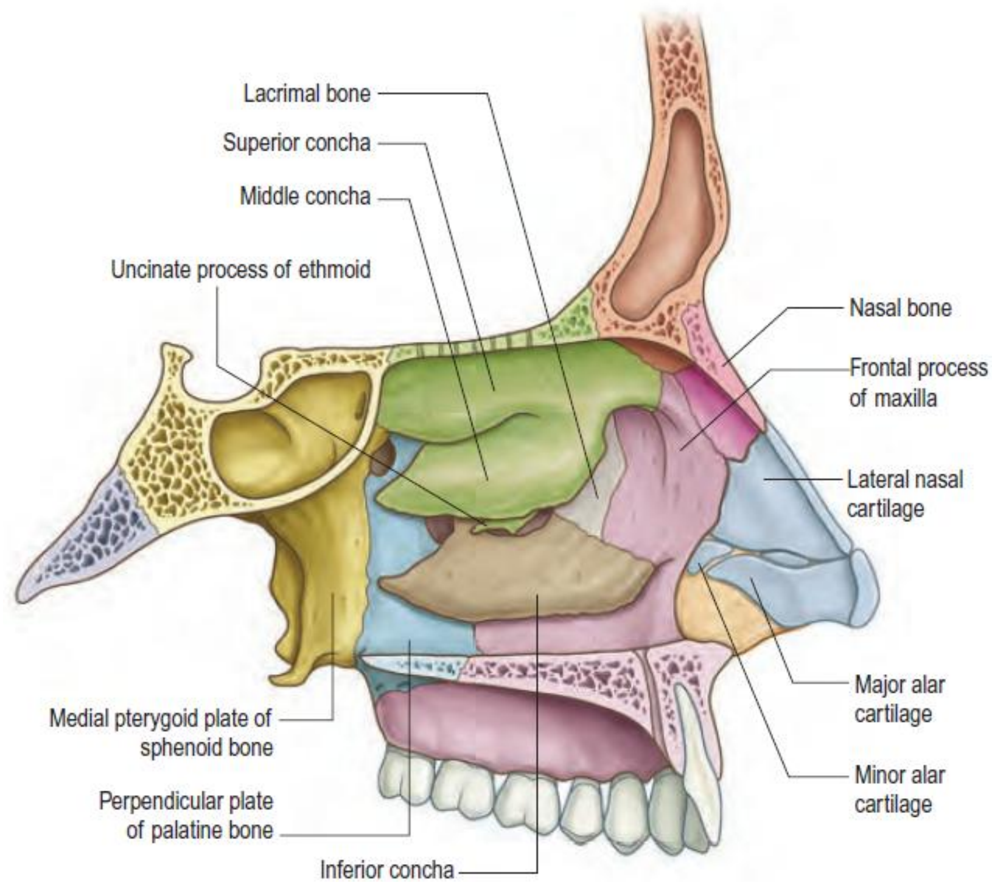


Fig. 39.5 A posterior view of the choanae.

Nasal Cavity Boundaries

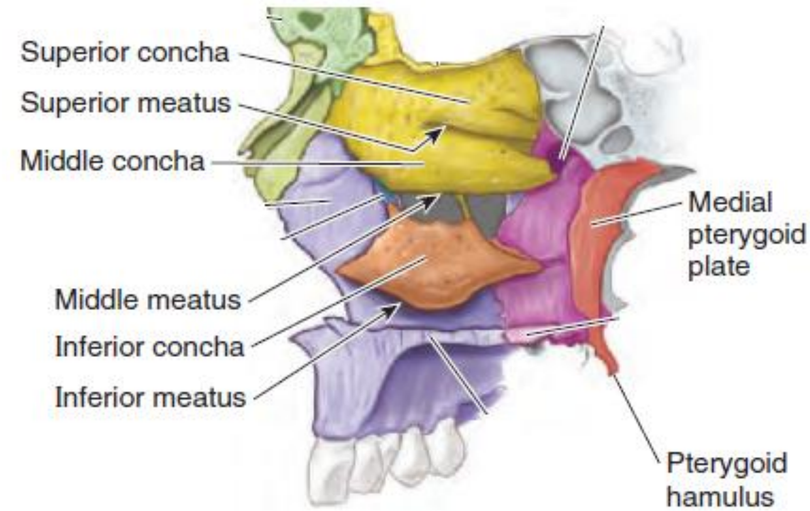
4 bony walls:

- Medial wall
  - **nasal septum**
- Roof: curved and narrow
  - nasal, frontal, ethmoid, sphenoid
- Floor: wider than roof
  - maxilla, palatine
- Lateral wall: irregular shape
  - nasal, ethmoid
  - maxilla, palatine
  - lacrimal, medial pterygoid plate of sphenoid bone
  - conchae

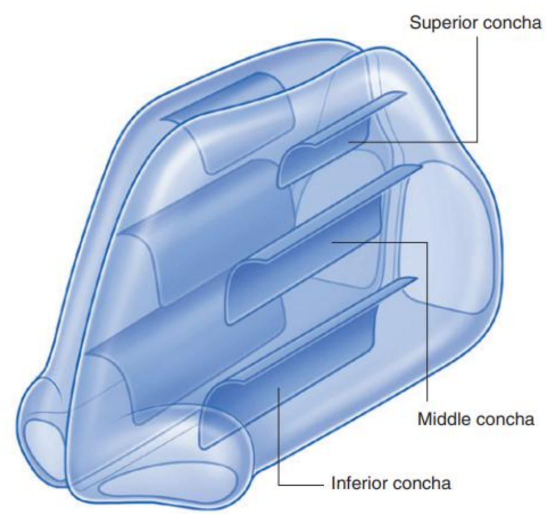


Conchae

- Nasal cavity lateral wall is raised into folds by three conchae
  - o convoluted bony structure offering greater surface area for heat exchange
  - o covered by mucus membrane containing large vascular spaces (ie. turbinate)
- Superior and Middle conchae
  - o ethmoid bone processes
- Inferior concha
  - o longest and broadest
  - o independent bone
- A meatus underlies each bony formation



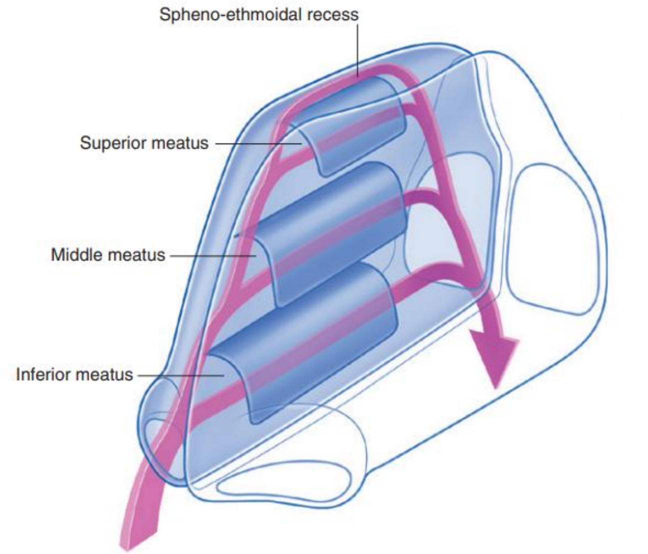
(A) Lateral wall of nasal cavity

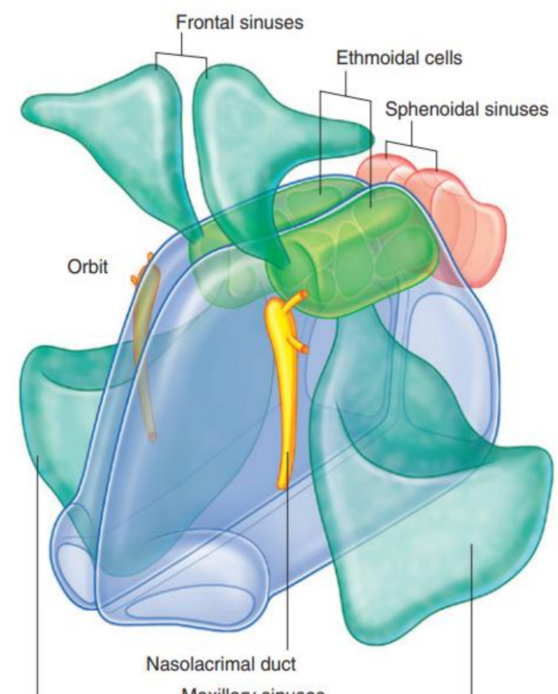


Meatuses

A meatus underlies each bony formation

- *Superior nasal meatus*
- *Middle nasal meatus*
- *Inferior nasal meatus*
- *Common nasal meatus*
- *Sphenoethmoidal recess*

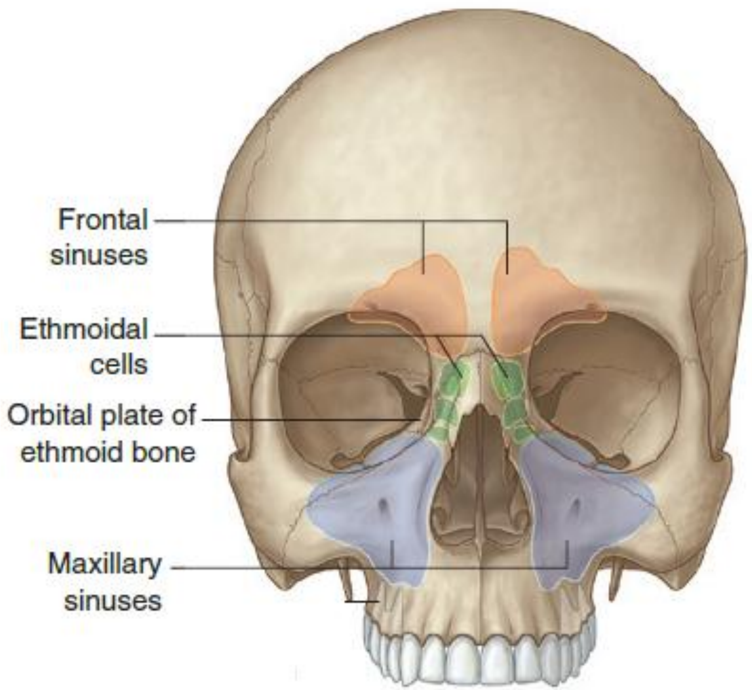


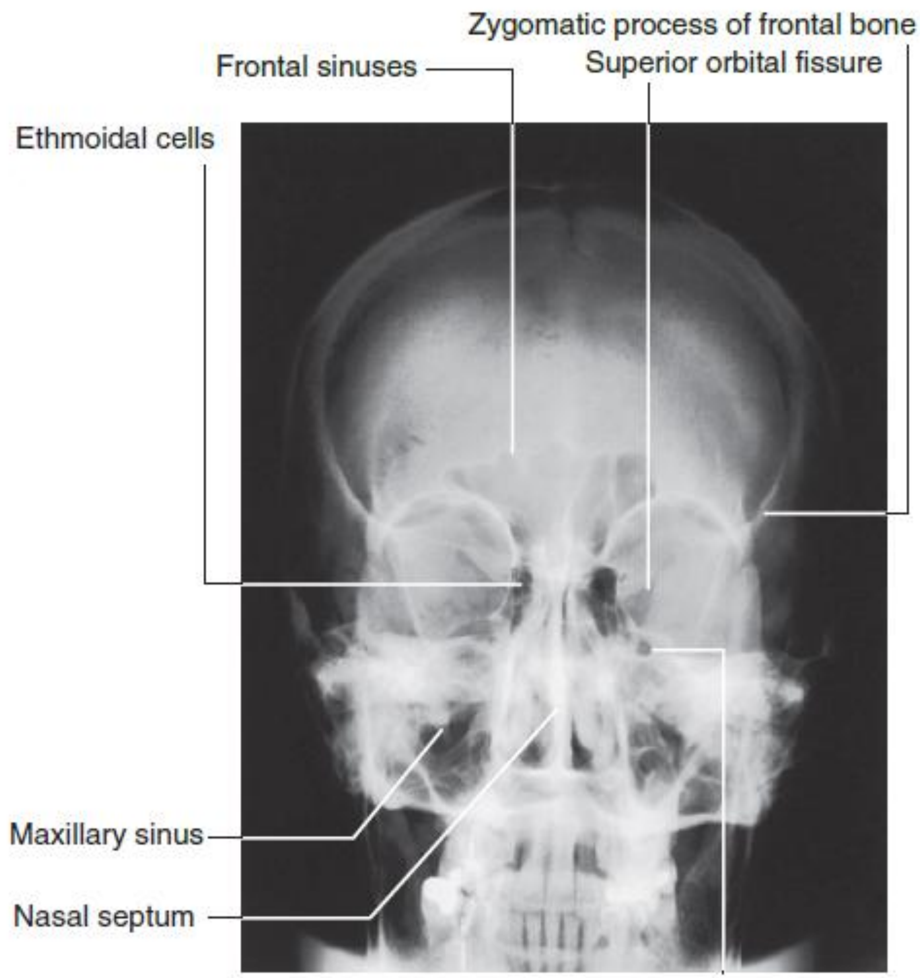


Paranasal Sinuses

Air-filled extensions of nasal cavity into the following cranial bones:

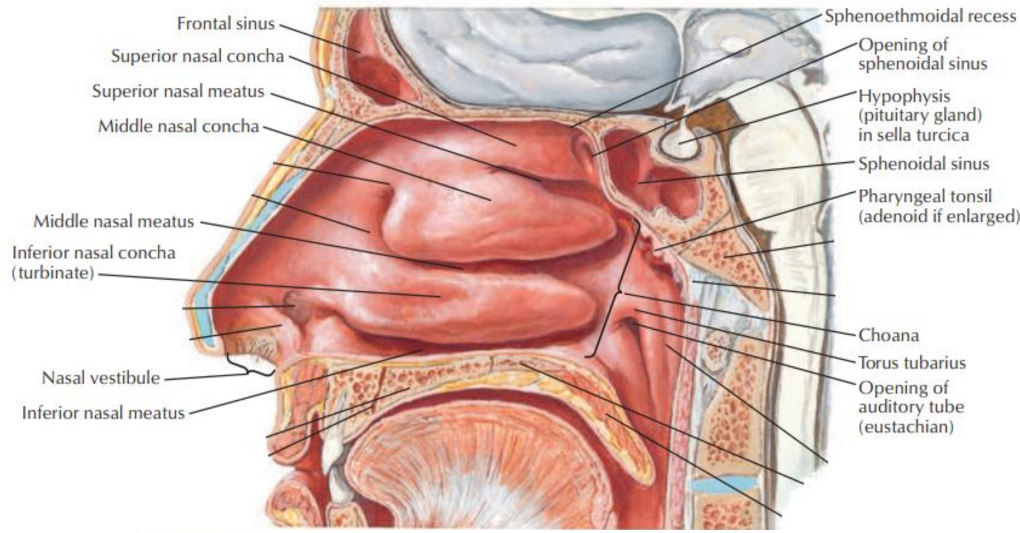
- Frontal
- Ethmoid
- Sphenoid
- Maxilla

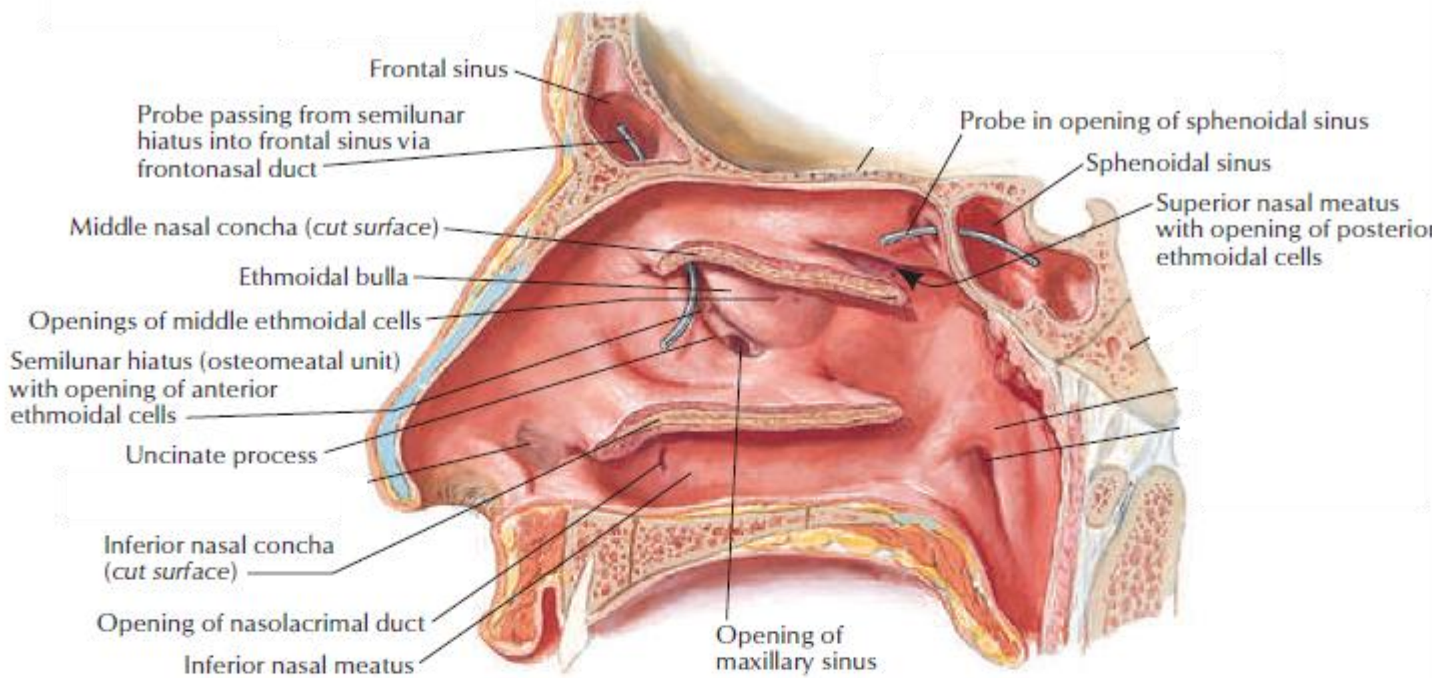




Nasal Cavity Connections

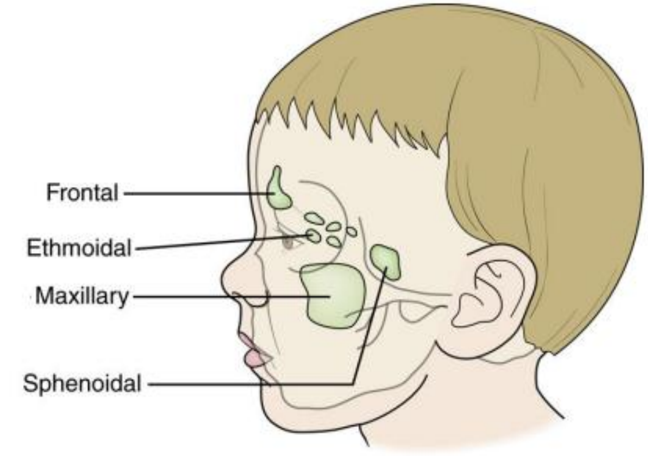
- Frontonasal ostia
- Ethmoid ostia
- Sphenopalatine foramen
- Maxillary sinus ostia
- Nasolacrimal ostia

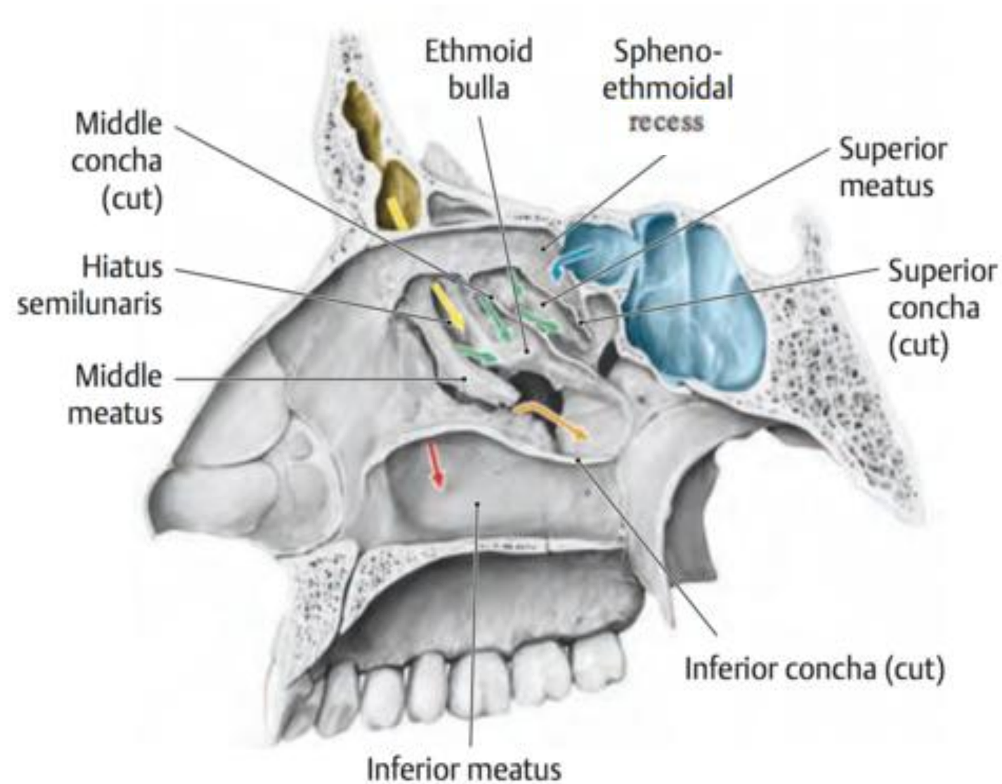




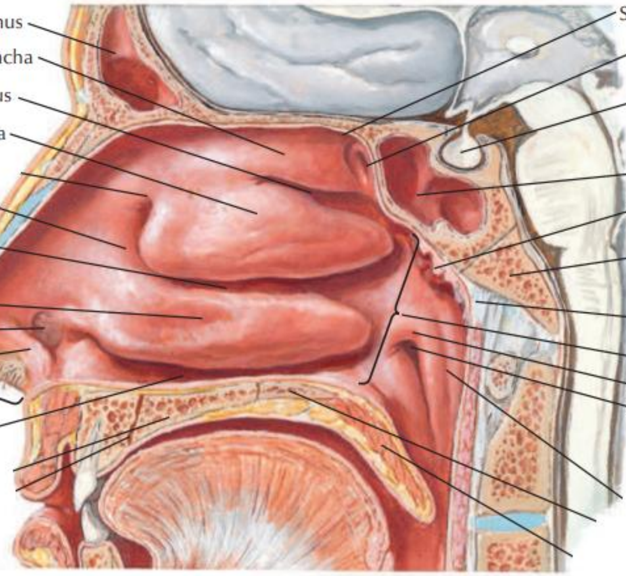
Nasal Cavity Passage Flow

- *Sphenoethmoidal recess*
    - receives sphenoidal sinus
- *Superior nasal meatus*
    - receives posterior ethmoid cells
- *Middle nasal meatus*
    - receives frontal and maxillary sinus, plus anterior and middle ethmoid cells
- *Inferior nasal meatus*
    - receives nasolacrimal duct
- *Common nasal meatus*

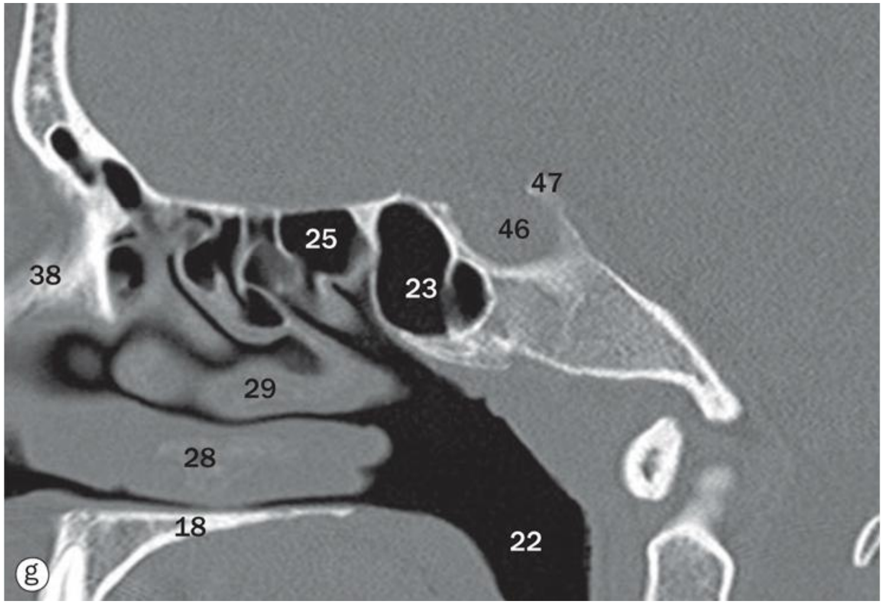




Nasal Cavity - Imaging



18: hard palate
22: nasopharynx
23: sphenoid sinus
25: posterior ethmoid air cells
28: inferior turbinate
29: middle turbinate
38: nasal bone
46: hypophyseal fossa



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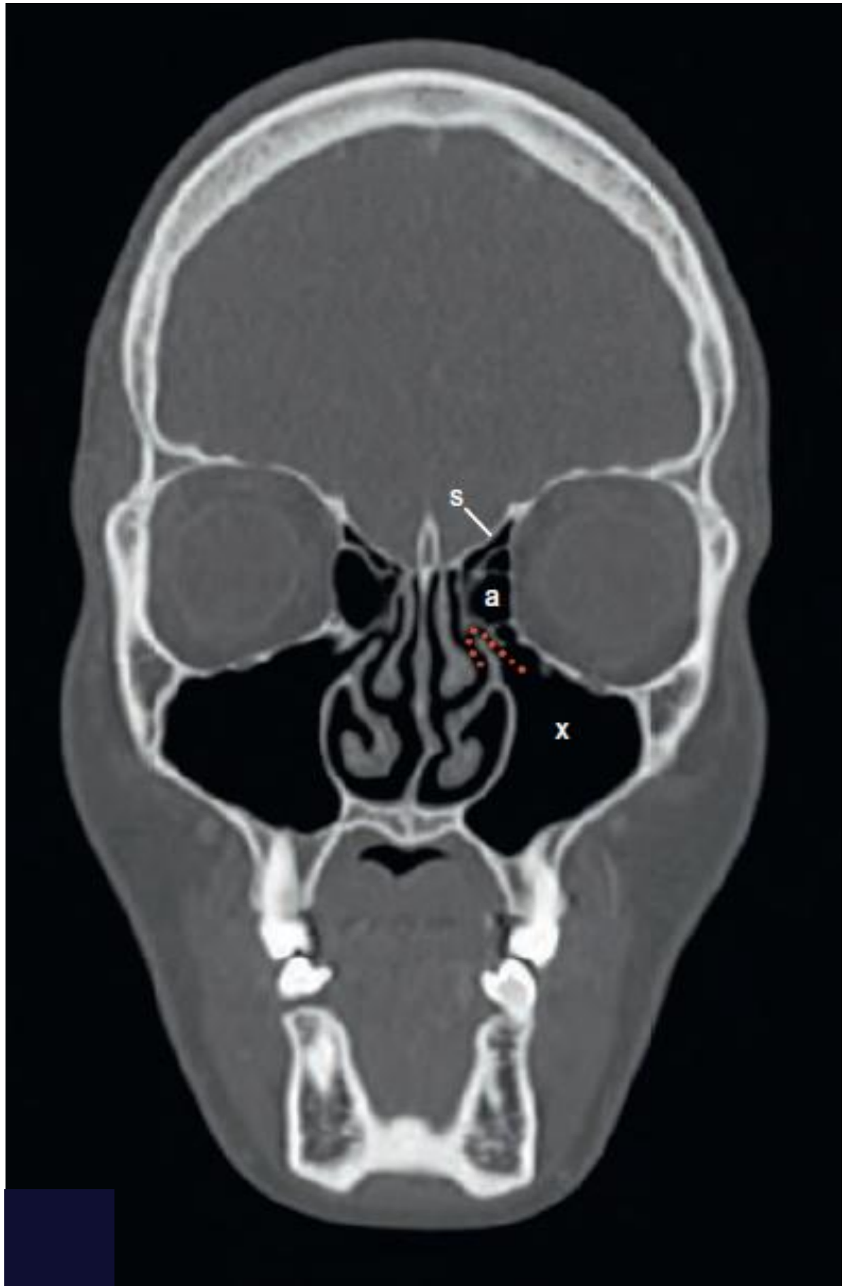
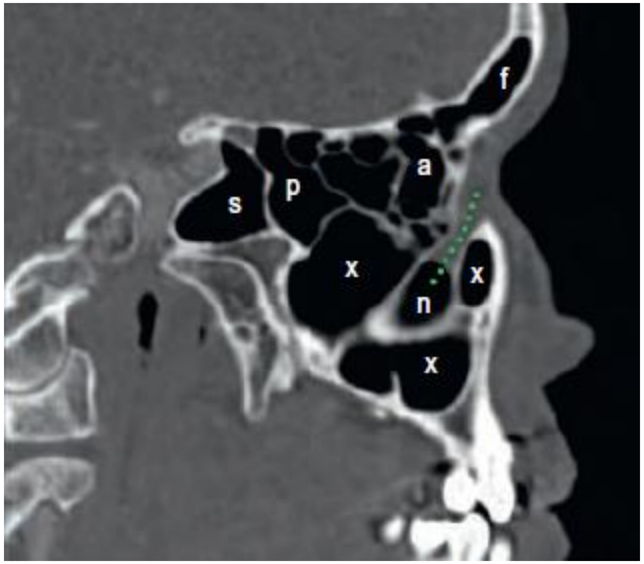
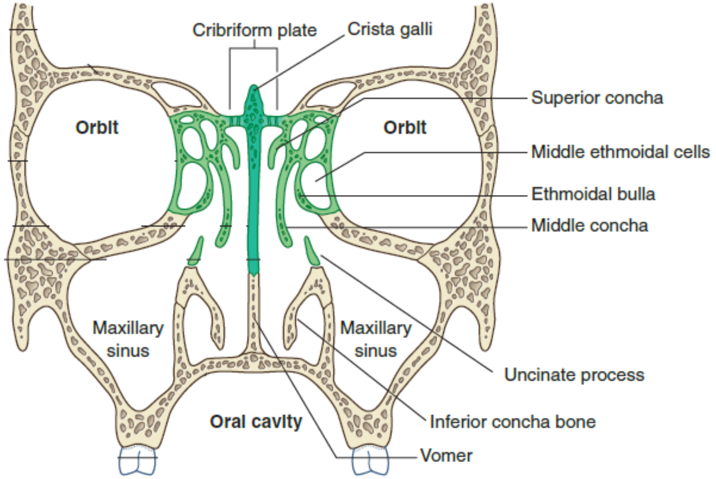
Air Pathways - Imaging

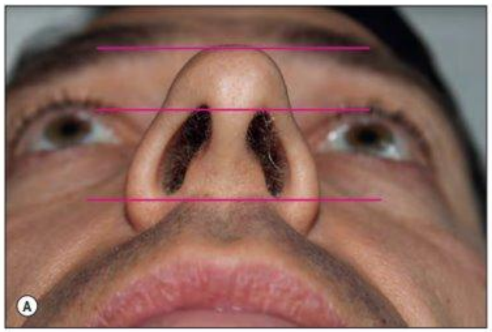


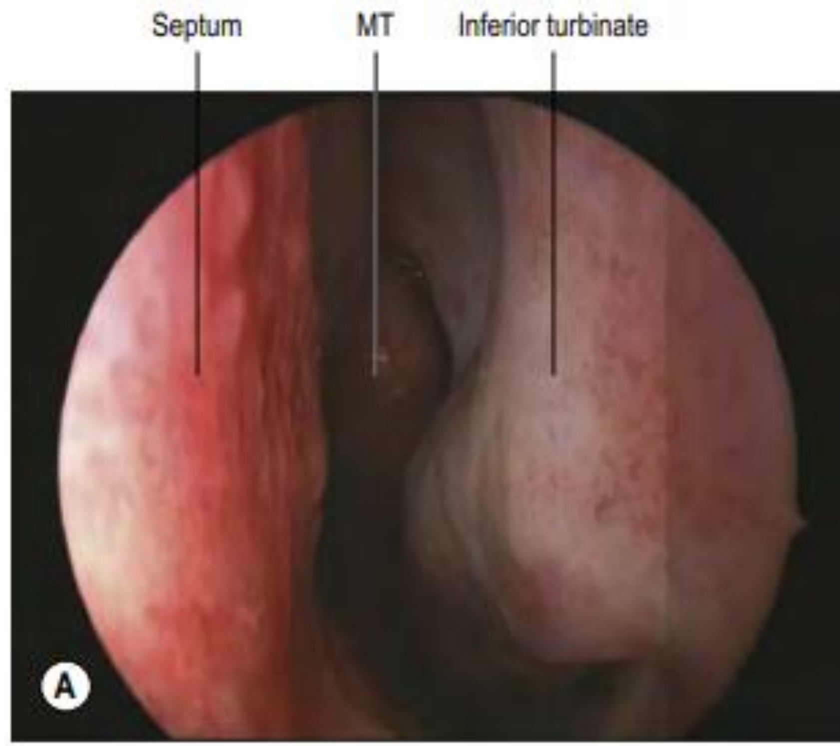


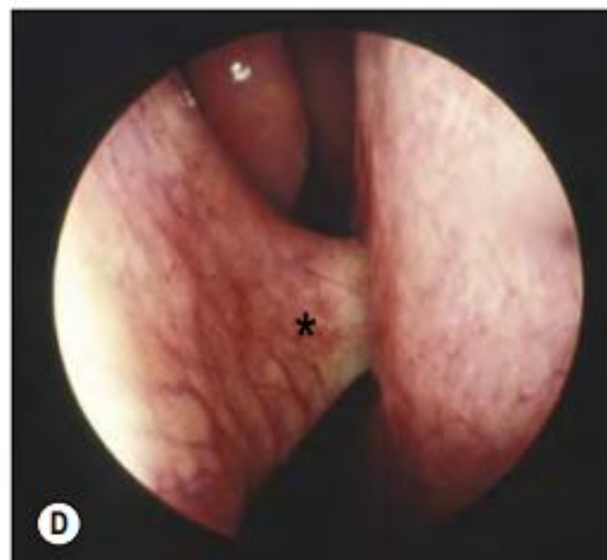
f: frontal air sinus
a: ant ethmoid air cells
p: posterior ethmoidal air cells
s: sphenoidal air sinus
x: maxillary air sinus
n: nasal cavity



Clinical Inspection

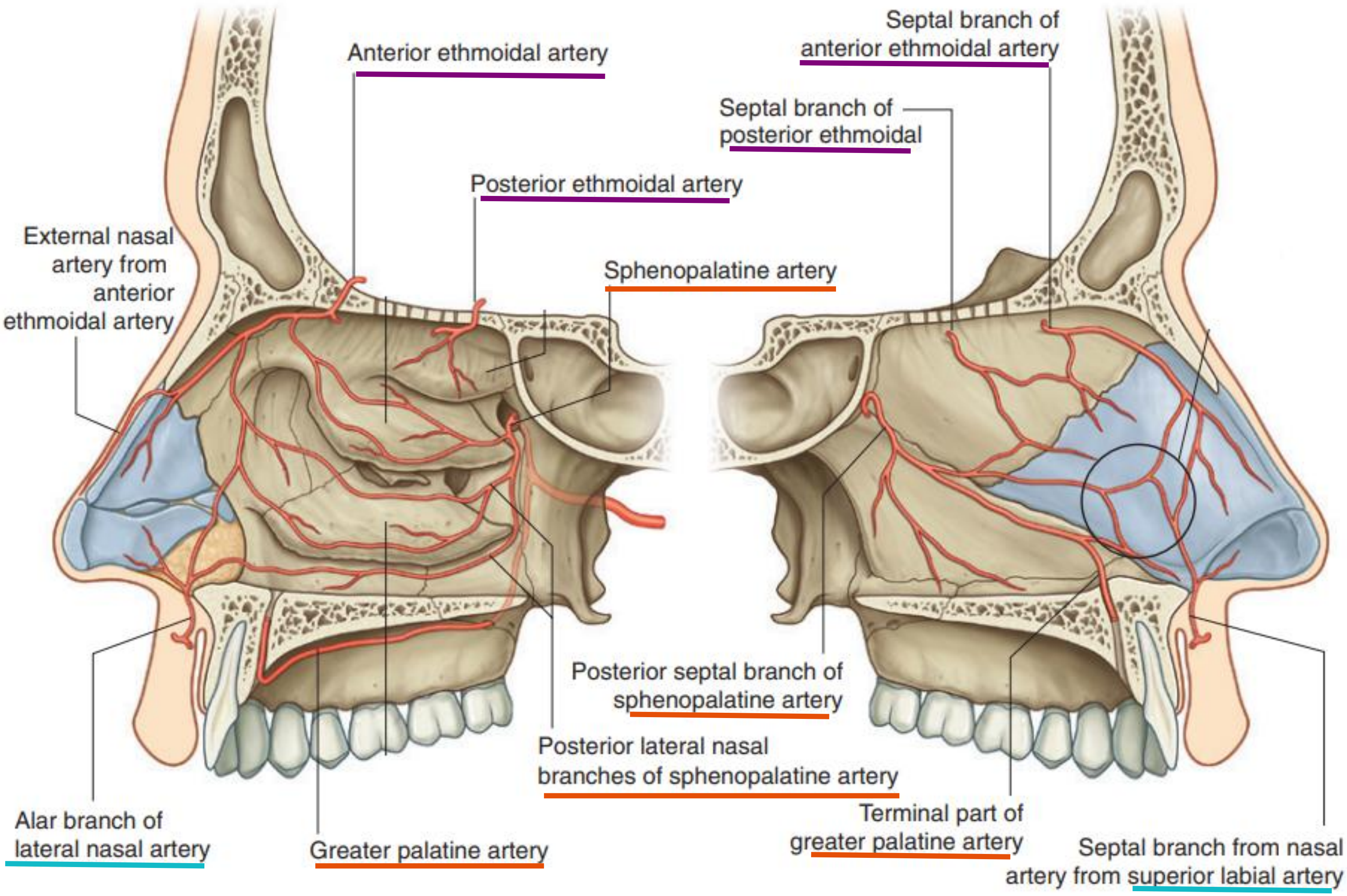






Nasal Vasculature

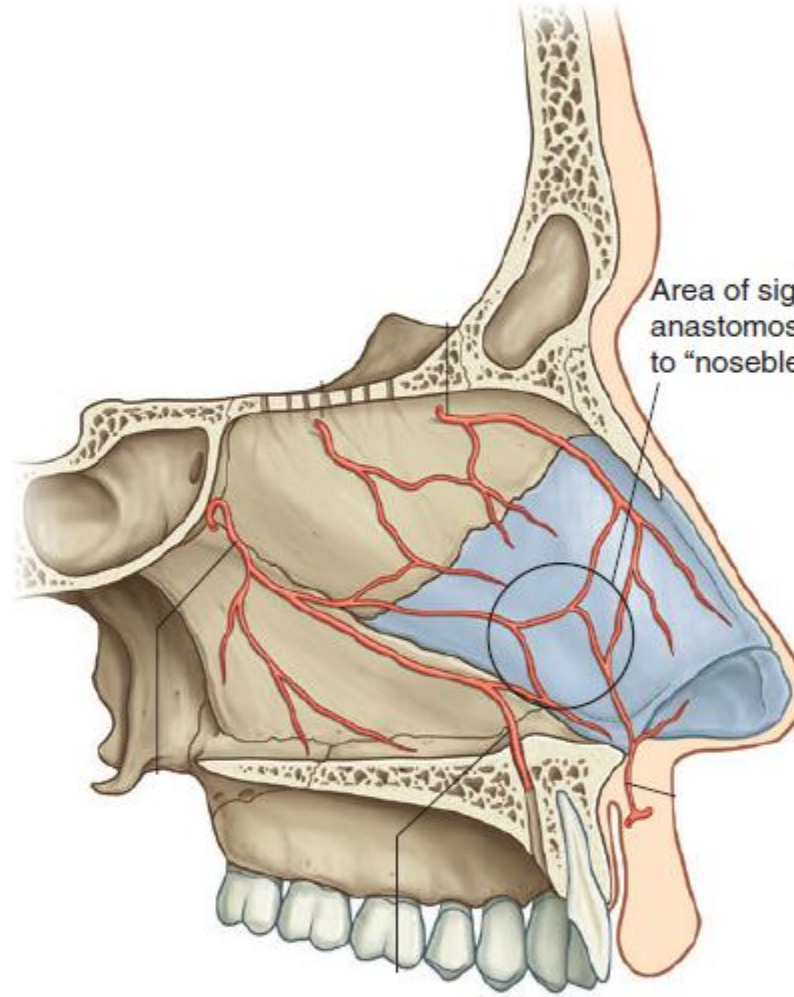
From:
Ophthalmic artery
Maxillary artery
Facial artery



Kiesselbach's Plexus

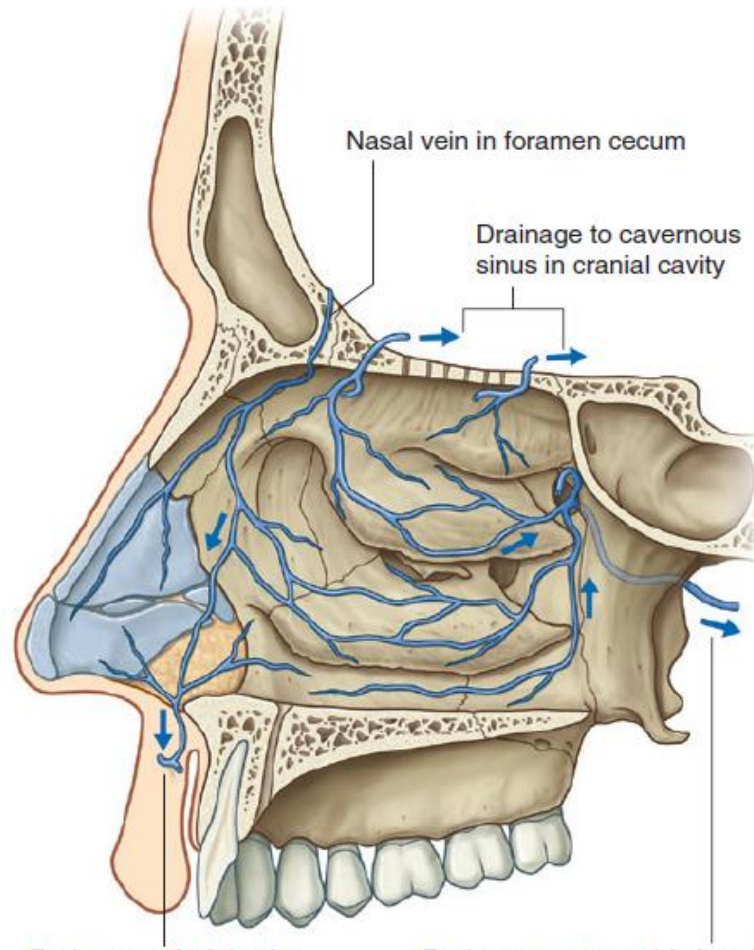
- The anterior part of the nasal septum contains a **highly vascularized** region of significant anastomoses
  - i.e. Little's area

- Most common site of significant **epistaxis** (nosebleed)



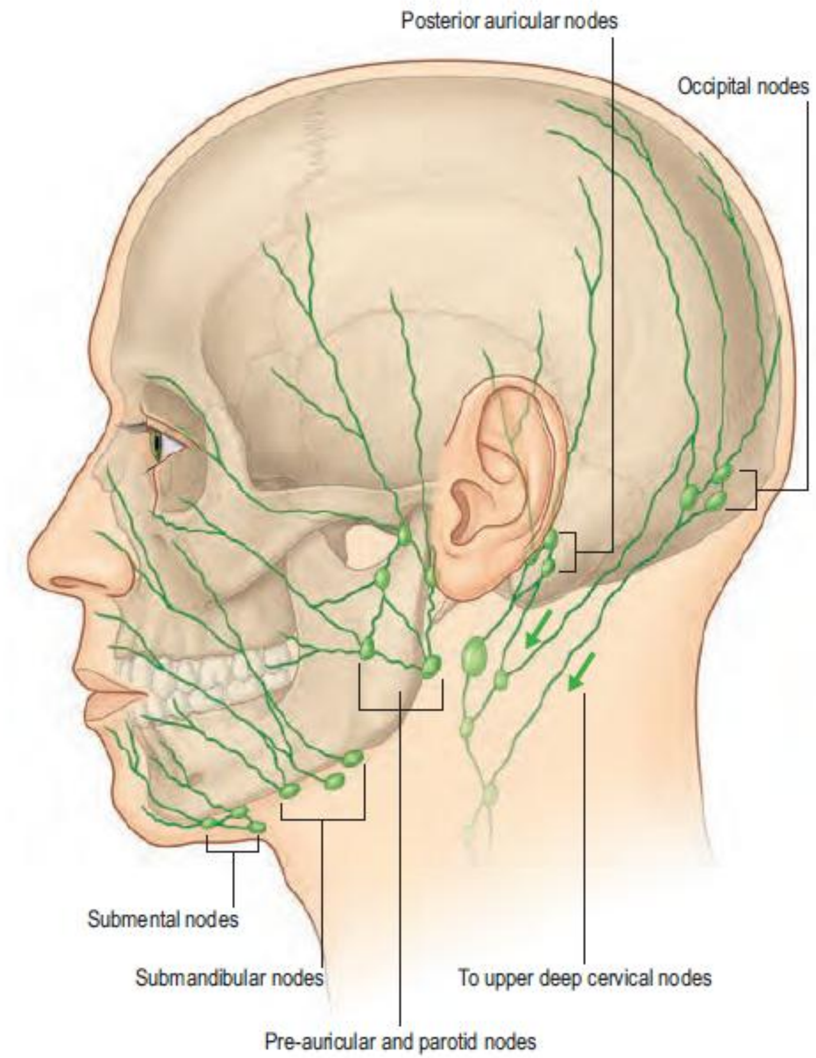
Venous Drainage

- Anterior nose
  - veins accompanying anterior ethmoidal veins, then pass to ophthalmic or *facial veins*
- Posterior nose
  - sphenopalatine vein, which drains into the *pterygoid venous plexus*
- Emissary veins
  - can connect to orbital surface of frontal lobe of brain



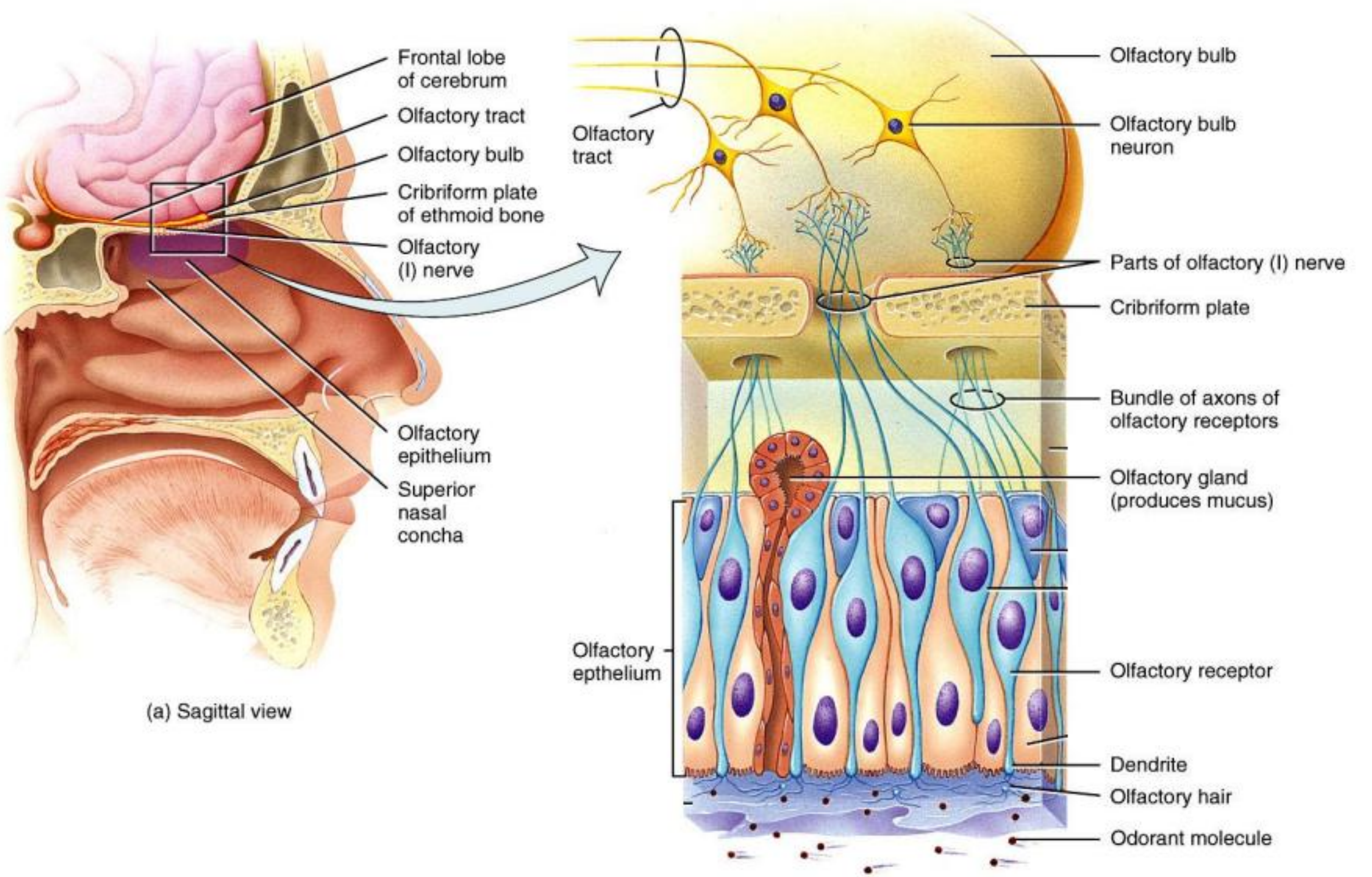
Lymphatic Drainage

- External nasal skin
  - submandibular nodes
  - Root of nose: **pre-auricular** and **parotid nodes**
- Anterior nasal cavity
  - end in submandibular nodes
- All drain into deep cervical nodes along the internal jugular vein
  - jugulo-digastric node
  - jugulo-omohyoid node



Olfactory Nerve

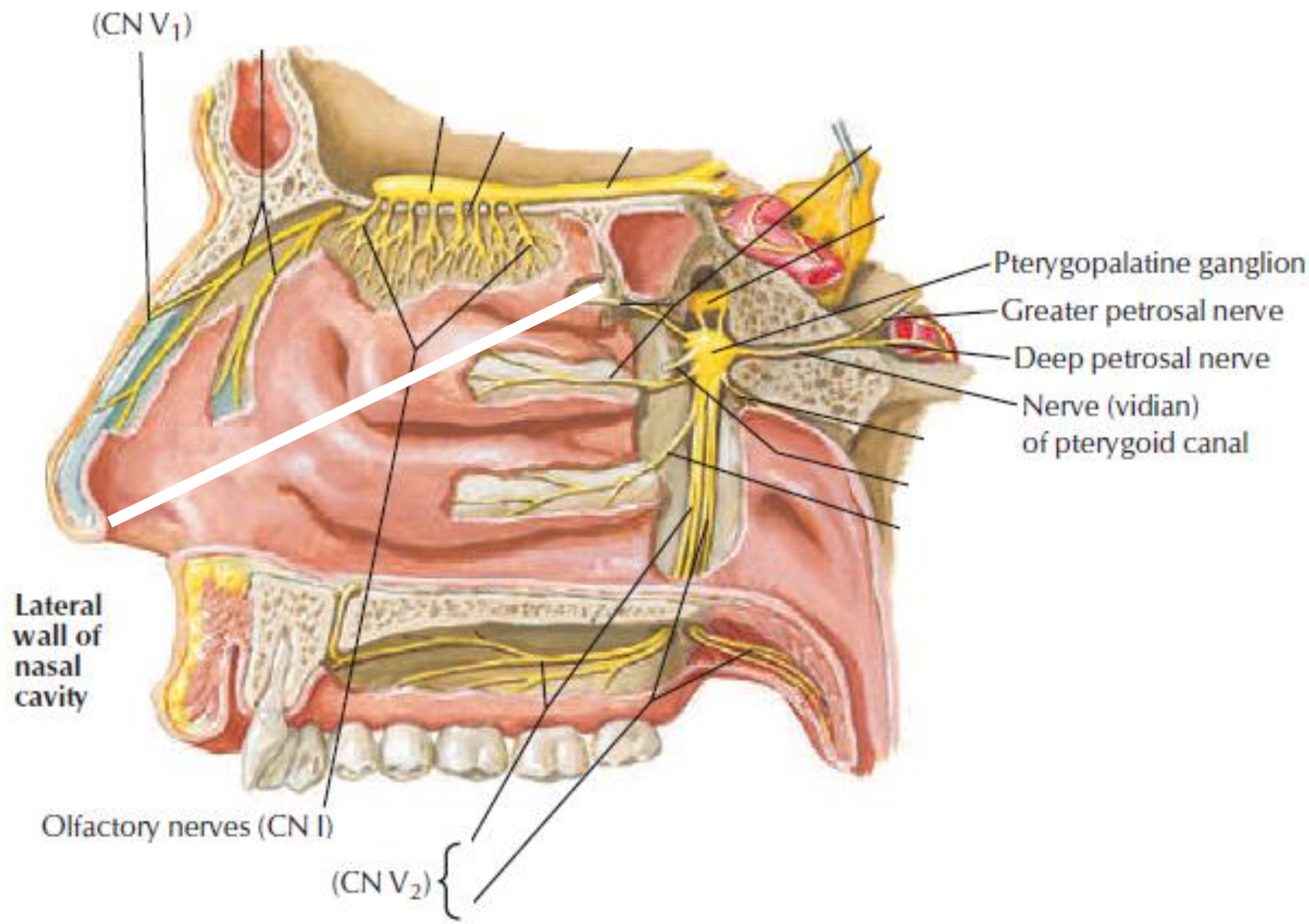
Special sensory bundles of olfactory nerve fibers (CN I) in the olfactory mucosa



Nasal Innervation

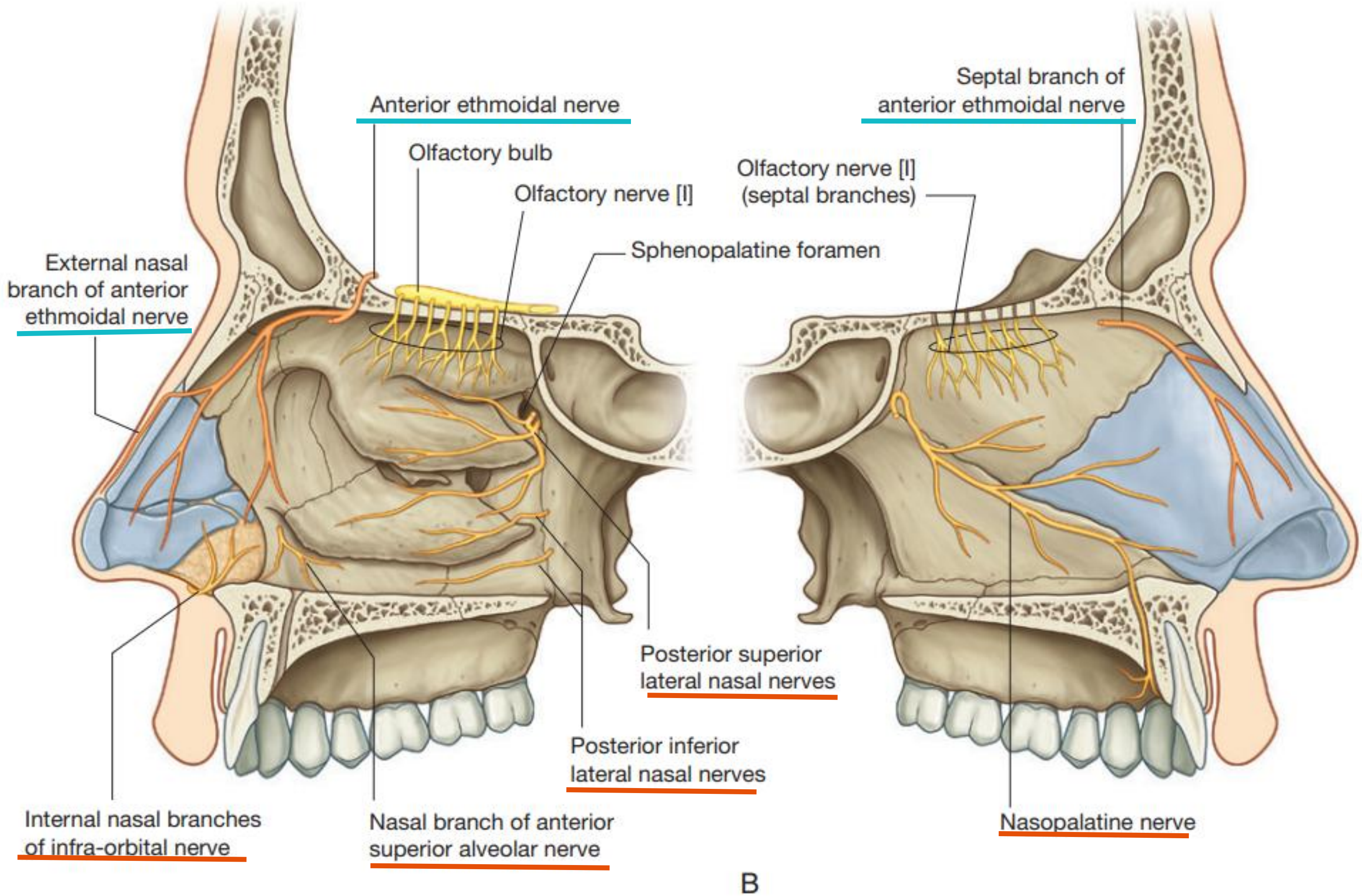
General guide:

- Anterosuperior: mostly V1
- Posteroinferior: mostly V2



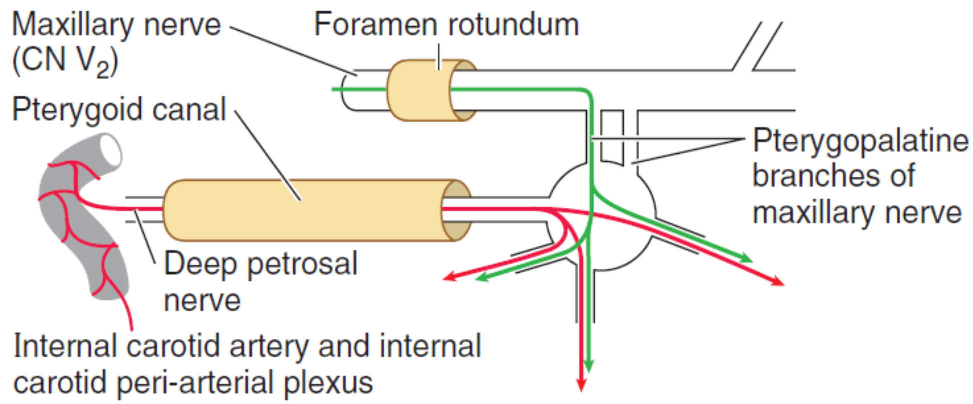
Nasal Innervation

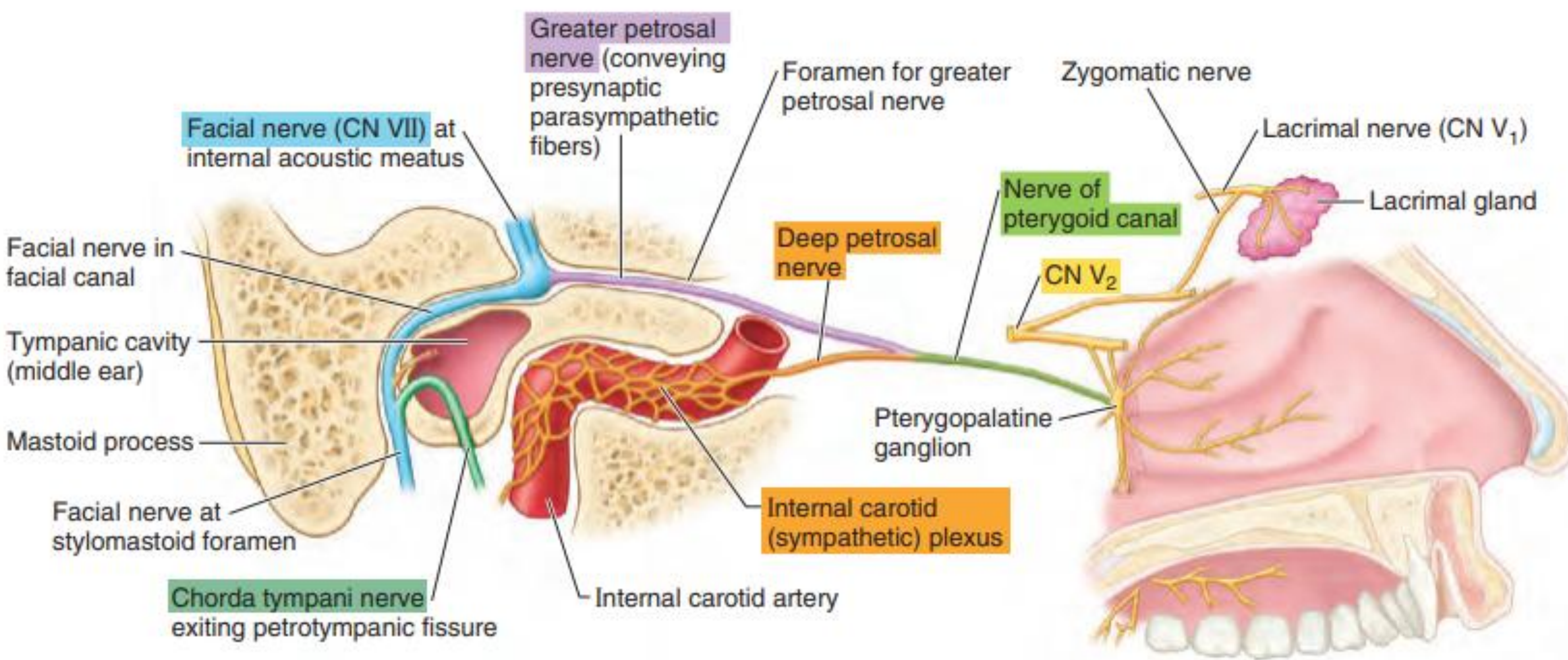
- Anterosuperior: mostly V1
- Posteroinferior: mostly V2



Autonomic Innervation

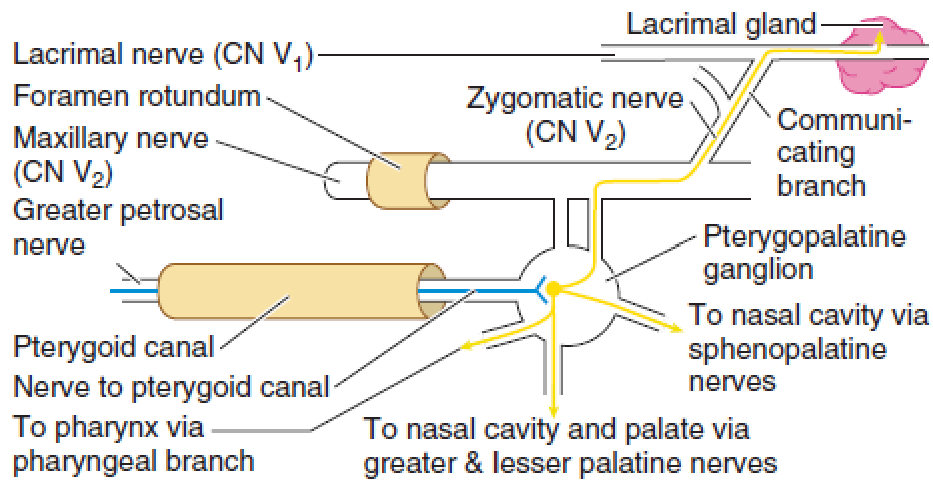
- Deep petrosal and greater petrosal fibers converge to form Vidian nerve
- Sympathetics pass without synapsing to nasal blood vessels

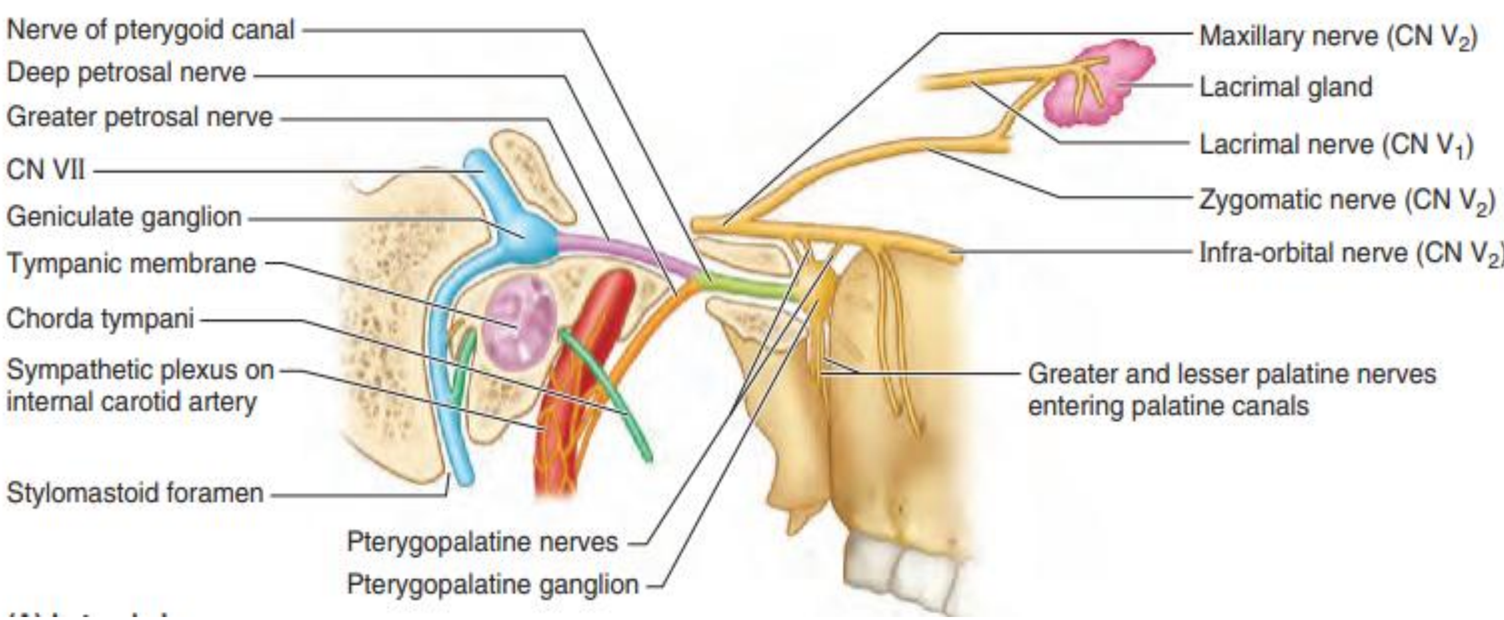




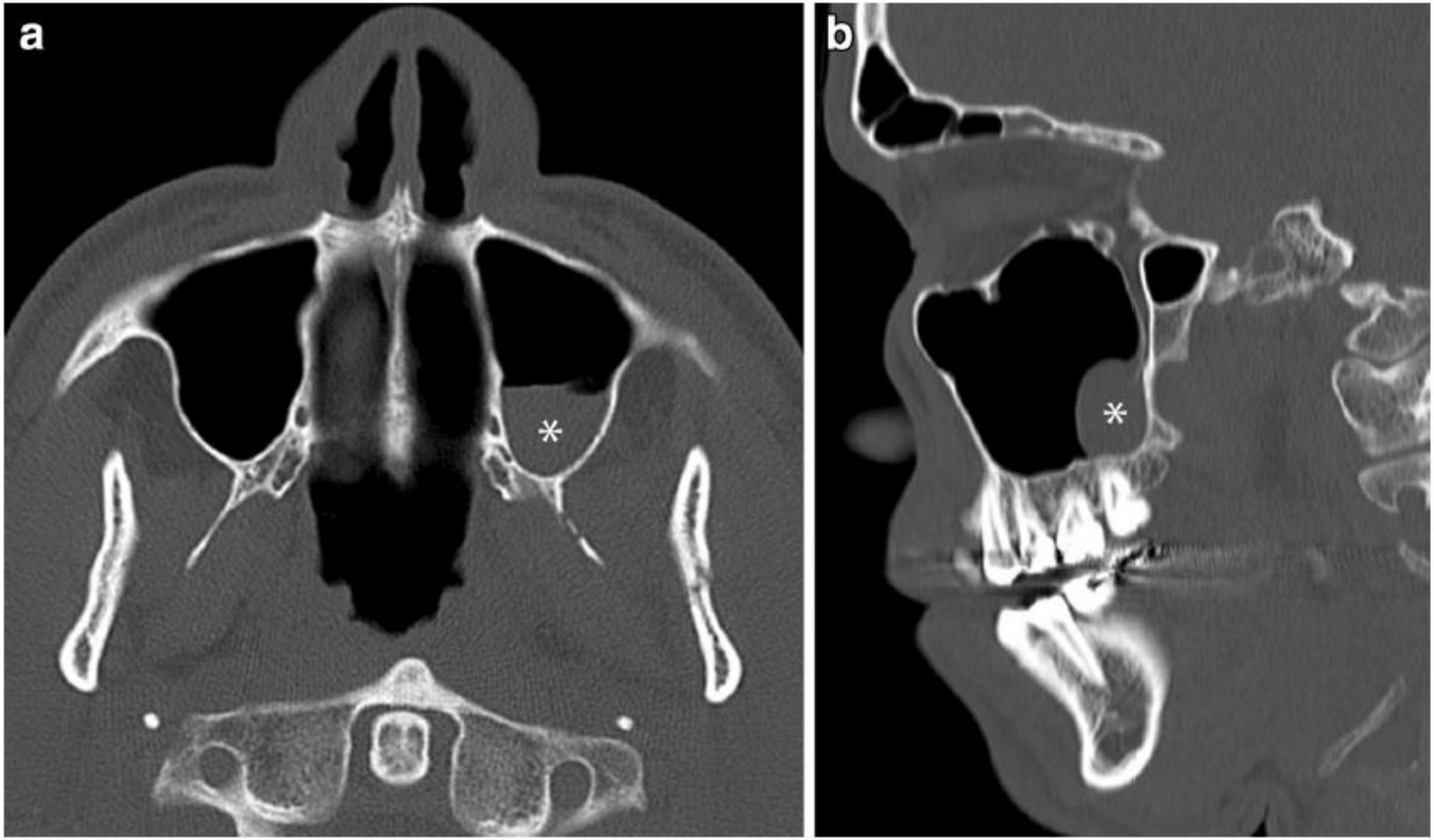
Autonomic Innervation

- Deep petrosal and greater petrosal fibers converge to form Vidian nerve
- Parasympathetics synapse to provide a secretomotor supply to the nasal mucus glands via branches of the maxillary nerves





Maxillofacial Image Interpretation



DOI: 10.1007/s10140-015-1343-4

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